

WEEK 1: Research & Strategic Discovery

Week 1 Objective

By the end of Week 1, you should:

1. Fully understand Oshi's product vision & monetization model
 2. Understand the current mobile architecture (DigitalOcean backend)
 3. Propose a clear web architecture decision (Render vs DigitalOcean)
 4. Understand the AI VTuber competitive landscape
 5. Deliver a short technical + product strategy memo
-

Technical Architecture Discovery (Days 1–2)

Audit Current System (Mobile + Backend)

Get familiar with our:

- Backend stack (language, framework)
- Hosting setup on DigitalOcean
- Database type
- Auth system
- API structure
- AI infra currently used

- Deployment process
-

Web App Architecture Decision

- Keep backend unified on DigitalOcean (same backend as mobile app)
 - Build web frontend on Vercel
 - Use same APIs for mobile + web
-

Tool Familiarization Requirements

Start using and build some starter projects using:

Claude Code

- How to use it for code generation
- How to structure prompts
- How to debug with it
- How to integrate an LLM
- How to design an algorithm

Deliverable:

Present your starter project built using Claude Code by the end of the week.

Vercel (Frontend Deployment)

We will use Vercel to deploy the front end. Get familiar with:

- Next.js deployment

- Environment variables
- Edge functions
- Preview deployments
- CI/CD workflows

Deliverable:

Deploy your basic starter app on Vercel.

PART 3 — Competitor Deep Dive (Days 4–5)

Research, use and present your findings on technical aspects of the following competitor apps:

- **IRIAM**
- **REALITY**
- **Kyodo**
- **VTube Studio**
- **OtherHalf**

PART 4 — AI VTuber Technical Research (End of Week 1)

Research:

Voice Cloning

- Open-source vs API options

- Latency benchmarks
- Real-time viability
- ElevenLabs is our preferred voice cloning tool, their new “ElevenLabs agents” are good for reducing latency of responses

LLM Integration

- Memory systems
- Personality conditioning (start thinking about our RAG system)
- Safety filters

Real-Time Avatar Rendering

- WebRTC?
- Unreal/Unity?
- Lightweight 2D animation?
- Browser compatibility?

Deliverable:

Architecture proposal for AI VTuber POC (1–2 pages)

Mandatory Outputs End of Week 1

In our Friday, you should present:

1. Live Vercel test deployment of your “Example app”
2. AI VTuber POC architecture proposal